

5(a)	all correct (1827) ;	1															
	<table> <tr> <th>phenotype</th><th>observed number</th><th>expected number</th></tr> <tr> <td>brown body colour, straight wings</td><td>2843</td><td>1827</td></tr> <tr> <td>brown body colour, curved wings</td><td>855</td><td>1827</td></tr> <tr> <td>black body colour, straight wings</td><td>842</td><td>1827</td></tr> <tr> <td>black body colour, curved wins</td><td>2768</td><td>1827</td></tr> </table>	phenotype	observed number	expected number	brown body colour, straight wings	2843	1827	brown body colour, curved wings	855	1827	black body colour, straight wings	842	1827	black body colour, curved wins	2768	1827	
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5(b)	any four from: 1 <i>ref. to</i> 3 degrees of freedom ; 2 the, χ^2 value / 2098, is greater than 7.815 (at $p=0.05$) ; 3 there is a significant difference ; 4 reject the null hypothesis ; 5 any difference is not due to chance / less than 5% (probability) that difference is due to chance ; 6 the, χ^2 value / 2098, is greater than, 11.345 at $p=0.01$ / 16.266 at $p=0.001$ or less than, 1% / 0.1% (probability), that difference is due to chance ;	4															

Question	Answer	Marks
5(c)	any four from: 1 greater numbers of parental phenotypes / fewer numbers of recombinants ; 2 genes are linked / autosomal linkage / genes are on same chromosome ; 3 alleles inherited together ; 4 B inherited with D and b inherited with d or B and D in the same gamete and b and d in the same gamete ; 5 no independent assortment ; 6 crossing over (produces recombinants) ; 7 during, meiosis / prophase I / gamete formation ; 8 AVP ; e.g. idea that genes are far enough apart on chromosome for recombination to occur	4

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